prometeia

Enterprise and Business line Risk Diagnostics

Status meeting 19.06.2026

Enterprise Risk Management

atb

AGENDA

Selected Deep Dives & Quick Wins

Internal And Trusted Customer Use Only Document Owned by Pro

Selected Deep Dives & Quick Wins

Stream Progress



	Activities	Deliverables	Timeline and assumptions
1 **Enterprise Risk Management – RAS Update**	➢ Review and update the current RAS metrics and definition of additional metrics to cover material risks (concentration, IRRBB, etc)	➢ Updated Risk Appetite Statement ➢ Methodology description for target RAS metrics	➢ 4-5 weeks starting from availability of required input ➢ The calculation of RAS metrics and their historical values will be provided by the Bank

- Completed

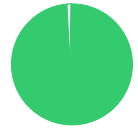
prometeia

all rights reserved

Internal And Trusted Partner Use Only Document owned by Prometeia

Selected Deep Dives & Quick Wins

Stream Progress



	Activities	Deliverables	Timeline and assumptions
2 **Retail Credit Risk - Collection process optimization**	➢ Extract delinquent portfolio data (DPD 1–45) including balance, product type, payment history ➢ Define a small set of collection prioritization indicators such as exposure size, DPD bucket, recent payment behavior and previous contact success. ➢ Develop a rule-based prioritization matrix assigning customers into priority tiers (High / Medium / Low).	➢ Call Prioritization Rule-set including a self-cure definition ➢ Collection performance dashboard (priority-based monitoring)	➢ 4-5 weeks starting from availability of required input ➢ Historical delinquency and payment data (customer-level) will be provided by the Bank.

Model development sample construction is completed.

Feature engineering activities are completed. 83 indicators are derived from the account and customer information tables.

Model development activites are completed.

Action matrix design completed.

prometeia

all rights reserved

Internal And Trusted Partner Use Only Document owned by Prometeia

Selected Deep Dives & Quick Wins

	Activities	Deliverables	Timeline and assumptions
3 **SME Credit Risk – Validate and fine-tune the current scorecard used for the credit decision making**	➢ Conduct a performance assessment, and identify potential refinements to scorecard structure and variable weighting	➢ Scorecard validation and fine-tuning, including assessment of the current scorecard design, indicative performance evaluation, and recommended adjustments to the scorecard structure, weights, and thresholds ➢ Guidance on how the enhanced scorecard should be embedded into the credit decision process	➢ 4-5 weeks starting from availability of required input ➢ The data required for the validation will be provided by the Bank. ➢ The fine-tuning will leverage the existing structure of the financial and qualitative assessments. The development of a structured scorecard for the financial module, including the definition of target variables, thresholds, and standard weights, will not be part of the project scope.

- ACB rating data and UW data are prepared and shared with Prometeia.
- Validation database constructed and the performance metrics calculated.
- **Fine tunings of the scorecard are ongoing.**

prometeia

all rights reserved

Internal And Trusted Partner Use Only Document owned by Prometeia

AGENDA

Retail Soft Collection Optimization

all rights reserved

prometeia

Internal And Restricted Use Only Document © Prometeia

Retail Soft Collection Optimization - Model Design

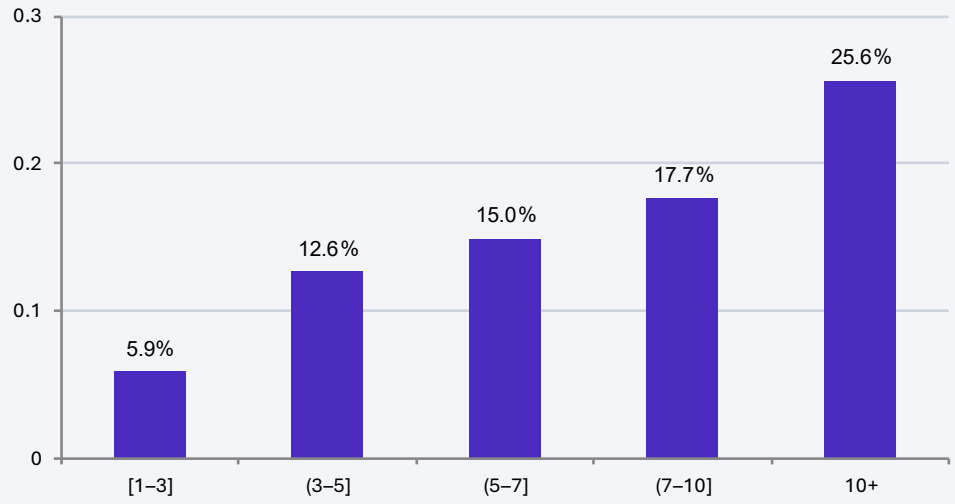
A single call strategy for the entire 1–45 DPD population would be sub-optimal. The proposed approach applies different decision logic by DPD band, while calibrating a common cross-segment priority queue for daily operations.

- **1–3 DPD** shows materially better bad-rate behavior; default action should be “wait / self-cure”.

- **4–10 DPD** is the main analytical intervention window; customers will be prioritized by expected roll-risk amount. There will be roll rate model to be developed for this segment which will estimate roll probability. Then the roll probability will be weighted by exposure to determine the call rank of the client.

- **10+ DPD** has higher observed bad rates and weaker incremental separability by risk drivers; a simpler exposure and DPD-severity ranking is more robust.

Roll Rate by DPD Buckets



5.9% — **1–3 DPD**

14.9% — **4–10 DPD**

25.6% — **10+ DPD**

ATB Monthly Data between 2024-Jan to 2025-Dec.
Bad Definition; having more than 30 DPD within next moth

prometeia

all rights reserved

Internal And Trusted Partner Use Only Document owned by Prometeia

Retail Soft Collection Process Optimization - Design

The solution combines segment-specific analytics with a unified operational queue. This preserves analytical fit by DPD band while allowing ATB to compare customers across bands under daily call-capacity constraints.

1 1–3 DPD

Severity / exposure ranking

- Favourable bad-rate profile
- Simple rule-based probability using max DPD within the last 3 months

Priority Score = P(roll) × EAD

2 4–10 DPD

Roll-risk prioritization

- Main intervention window
- Model: tree-based regression model to predict the rolling probability to the 30+ delinquency bucket within one month
- **Action; call based on the priority score**

Priority Score = P(roll) × EAD

3 11– 30 DPD

Severity / exposure ranking

- Bad rate already high
- Risk differentiation is limited
- Rule-based ranking is more robust

Score = EAD × DPD roll rate

Calibration principle	10+ DPD customers should receive higher rank on average, but should not automatically outrank every 1–10 DPD customer. A high-EAD, high-roll-probability 4–10 DPD customer may represent higher expected risk than a low-EAD 11–45 DPD customer.

prometeia

all rights reserved

Internal And Trusted Partner Use Only Document owned by Prometeia

Retail Soft Collection Process Optimization

4-10 DPD Segment – Model Development Approach

A dedicated roll-rate prioritization model was developed for the 4–10 DPD population. The model assesses the likelihood of rolling into the 30+ delinquency bucket within the following month using behavioral delinquency indicators with statistical decision tree algorithm.

1 Sample Construction

- Monthly customer-level observation framework was constructed using account and customer behavioral data
- Jan-2024 to Sep-2025 was used as the development sample, while October 2025 to December 2025 was reserved for out-of-time validation.
- Target was defined as rolling into the 30+DPD bucket within the following month

2 Feature Engineering & Univariate Analysis

- 77 behavioral indicators were derived using demographics, delinquency, payment behavior and exposure dynamics
- Variables with univariate GINI values below 0.05 were excluded from the initial longlist
- Variables with correlation levels above 0.70 were grouped together and variables with higher univariate GINI values were selected within each group
- The final shortlist was reduced to 16 variables for decision tree model development

3 Model Development

- Multiple decision tree structures were developed using different shortlisted variable combinations and hyperparameter settings in order to identify the most stable and interpretable segmentation structure
- Model candidates were evaluated based on discriminatory power, variable diversity and interpretability.
- The final decision tree structure was fine-tuned to improve business interpretability and operational usability

all rights reserved

prometeia

Internal And Trusted Partner Use Only Document owned by Prometeia

Retail Soft Collection Process Optimization

Model Structure: 4-10 DPD segment

The final roll-rate model was consisting of 5 indicators and a decision-tree-based segmentation structure. The resulting framework consists of 8 final risk segments with differentiated roll-rate probabilities, achieving 36.6% Train GINI and 40.4% Out-of-Time GINI performance.

Final Model Variable	Description
DPD_MAX_MAX12	Maximum observed DPD within the last 12 months
DPD_RATIO_ADJ_MAX12	Maximum delinquent exposure ratio observed within the last 12 months
DPD_RATIO_ADJ_V3LM	Difference between the latest month delinquent exposure ratio and the maximum delinquent exposure ratio observed during the previous 3-month period
DPD30_ADJ100_CNT_L6M	Number of months with 30+ DPD occurrence within the last 6 months
DPD_RATIO_ADJ_VLM	Latest month delinquent exposure ratio

The final segmentation structure shows clear bad-rate separation across the final nodes, with roll rates increasing from 8% to 42% across the risk hierarchy. In addition, train and out-of-time samples show similar bad-rate behavior within each segment, supporting the stability of the proposed framework.

Risk Class	Train Obs%	Train Bad Rate	OOT Obs %	OOT Bad Rate	Roll Probability	Rule
Class 1	40%	8%	45%	8%	8%	DPD_MAX_MAX12 ≤ 7 **AND** DPD_RATIO_ADJ_MAX12 > 0.025
Class 2	20%	11%	19%	10%	11%	7 < DPD_MAX_MAX12 ≤ 30 **AND** DPD_RATIO_ADJ_MAX12 > 0.03
Class 3	12%	15%	8%	9%	14%	DPD_MAX_MAX12 ≤ 30 **AND** DPD_RATIO_ADJ_MAX12 ≤ 0.025
Class 4	4%	16%	3%	17%	16%	7 < DPD_MAX_MAX12 ≤ 30 **AND** 0.025 < DPD_RATIO_ADJ_MAX12 ≤ 0.03
Class 5	7%	21%	9%	23%	22%	DPD_MAX_MAX12 > 30 **AND** DPD_RATIO_ADJ_V3LM > -0.4 **AND** DPD_30_ADJ100_CNT_L6M ≤ 1 **AND** DPD_RATIO_ADJ_VLM ≤ 0.2
Class 6	7%	27%	6%	28%	28%	DPD_MAX_MAX12 > 30 **AND** DPD_RATIO_ADJ_V3LM > -0.4 **AND** DPD_30_ADJ100_CNT_L6M ≤ 1 **AND** DPD_RATIO_ADJ_VLM > 0.2
Class 7	5%	35%	5%	39%	36%	DPD_MAX_MAX12 > 30 **AND** DPD_RATIO_ADJ_V3LM > -0.4 **AND** DPD_30_ADJ100_CNT_L6M > 1
Class 8	5%	41%	5%	44%	42%	DPD_MAX_MAX12 > 30 **AND** DPD_RATIO_ADJ_V3LM <= -0.4

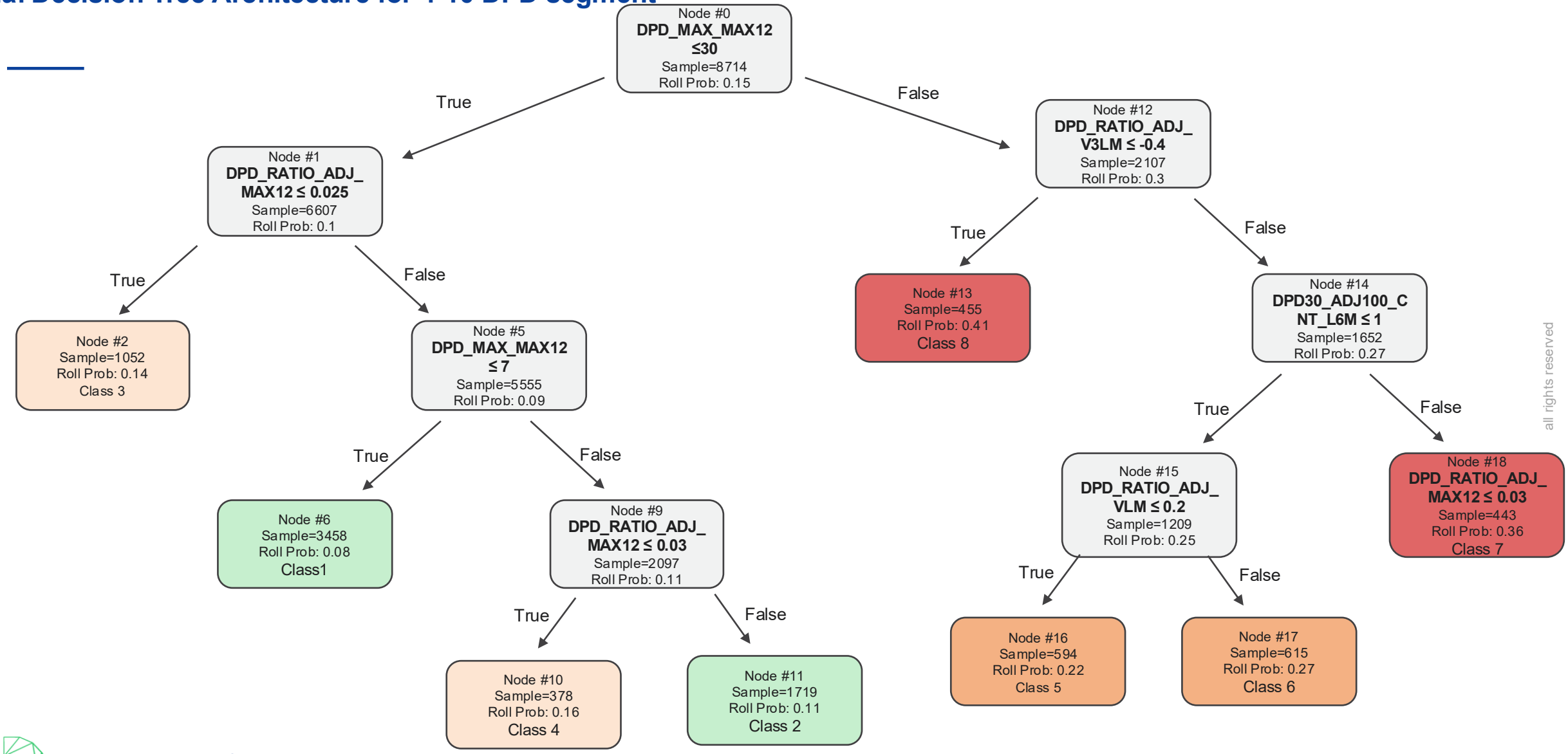
all rights reserved

prometeia

Internal And Trusted Partner Use Only Document owned by Prometeia

Retail Soft Collection Process Optimization

Final Decision Tree Architecture for 4-10 DPD segment



all rights reserved

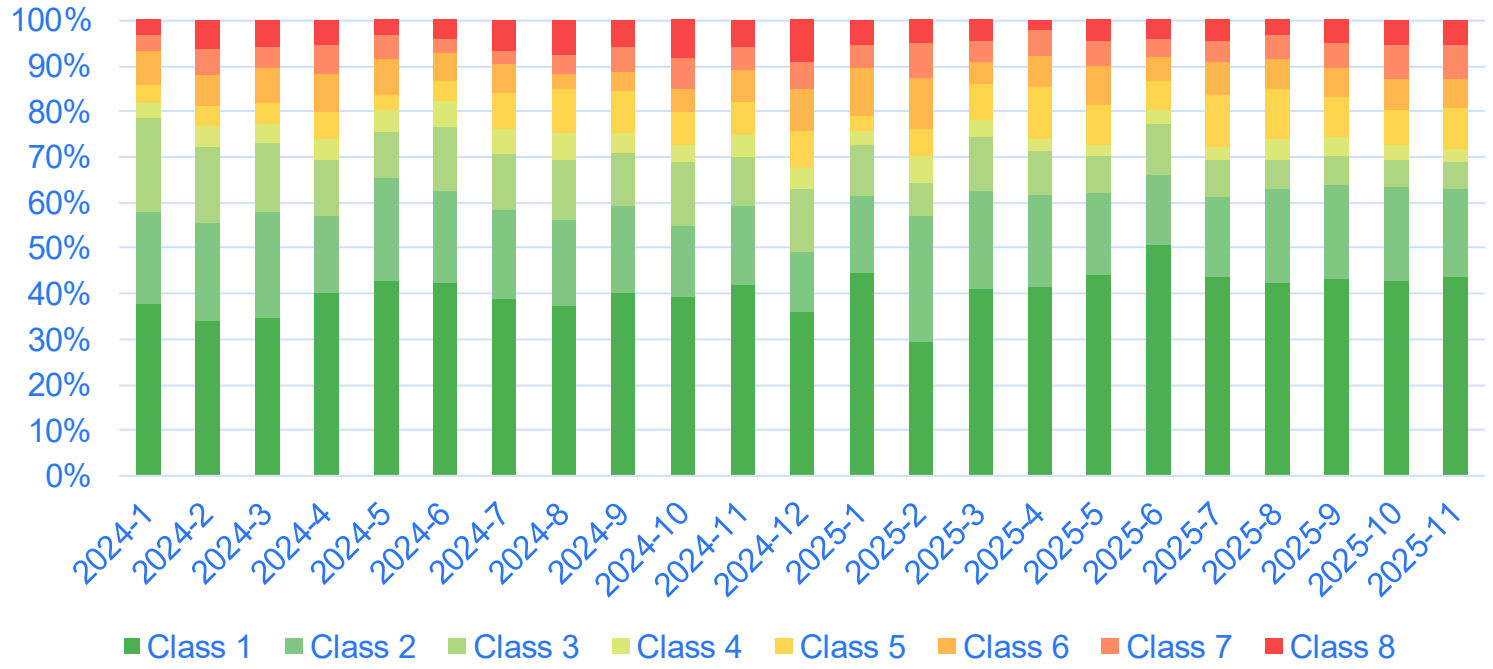
prometeia

Internal And Trusted Partner Use Only Document owned by Prometeia

Retail Soft Collection Process Optimization

Monthly Analysis of Model Risk Classification

The portfolio distribution across risk classes remains broadly stable over time, with no material concentration shift observed toward higher-risk segments throughout the monitoring period.



all rights reserved

prometeia

Internal And Trusted Partner Use Only Document owned by Prometeia

Retail Soft Collection Process Optimization

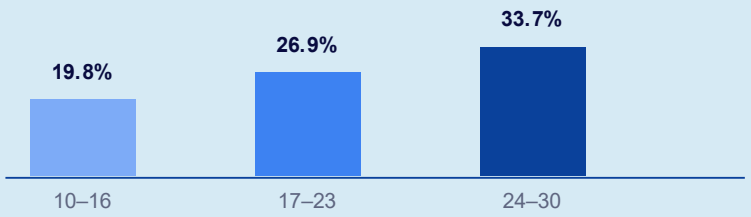
Priority Score Assignment for 10+ DPD Segment

Empirical roll-rate evidence for 10+ DPD buckets

Observed deterioration is monotonic across DPD buckets; therefore the observed bucket-level rate can be used directly as the probability input.

DPD Bucket	OBS	OBS %	BAD	Roll Rate
[10,17)	5,541	41.4%	1,098	**19.8%**
[17,24)	4,530	33.8%	1,219	**26.9%**
[24,30]	3,318	24.8%	1,117	**33.7%**

DPD-based Roll Rate



Scoring logic: use the roll rate directly

10+ DPD Priority Score

DPD Bucket Roll Rate × Total Exposure

The observed roll rate is treated as the probability component of the expected-risk score.

Cross-segment comparability

4–10 DPD: Customer-level Roll Probability × Total Exposure

10+ DPD: DPD Bucket Roll Rate × Total Exposure

all rights reserved

prometeia

Internal And Trusted Partner Use Only Document owned by Prometeia

Cost-based action matrix calibration for the 1–30 DPD portfolio

Translate the analytical roll-risk framework into an economically justified daily treatment strategy by comparing customer-level expected roll amount with the operational cost and expected uplift of each collection action.

Expected Roll Amount

ERA = Roll Probability × Total Exposure

Break Even ERA

Action Unit Cost / Uplift Factor

Action Decision

Feasible if ERA > Break even ERA

Probability sources

- 1–3 DPD: simple rule-based probability using max DPD within the last 3 months
- 4–10 DPD: model-based roll probability
- 11–30 DPD: DPD-bucket roll rate

*The break-even ERA formula converts each collection action into a minimum expected roll amount threshold. The logic is simple: an action is economically justified only if the expected benefit generated by the action exceeds its operational cost. Since the action is assumed to reduce only a fraction of the expected roll amount, the break-even ERA is calculated as action cost divided by the assumed uplift. Therefore, higher-cost channels require higher expected roll amount, while stronger uplift assumptions reduce the break-even threshold.

all rights reserved

prometeia

Internal And Trusted Partner Use Only Document owned by Prometeia

Collection Cost Inputs

0.031
AZN per domestic processing SMS

None
Current IVR / robo call channel

6
Early delinquency agents

219
Daily average call attempts per agent

107
Daily worked customers per agent

82
Daily reached customers per agent

76%
RPC

Manual call cost input

Input	Value	Interpretation
PKİM cost	184,547 AZN / 4 months	Total unit cost pool provided by ATB
Employees	15	Used to derive monthly cost per employee
Monthly cost per employee	3,076 AZN	Fully-loaded proxy for agent cost
Daily average call attempts	219	A client is called more than once in a day
Daily worked customers	107 customers	Number of distinct customers called in a day per agent
Daily reached customer	82 customers	Number of customers with successful contact (RPC 76%)
Current IVR / robo call	Not available	Included later as optional future-state channel

all rights reserved

Internal And Trusted Partner Use Only Document owned by Prometeia

Derived unit costs for the action matrix

Monthly cost per agent
3,076 AZN

Daily cost per agent
3,076 / 22 working days = 140 AZN

Cost per customer-day
140 / 80 = 1.75 AZN

Cost metric	Estimated unit cost	Use in action matrix
Initial SMS	0.031 AZN	Baseline SMS sent to all newly delinquent customers
Reminder SMS	0.031 AZN	Incremental low-cost reminder after initial SMS
Robo call / IVR attempt	~0.20 AZN	Illustrative future-state option; not currently available
Manual customer-day	~1.75 AZN	Preferred cost reference for customer-level manual follow-up

Key implication

SMS costs are economically negligible compared with manual calling.

The 1.75 AZN customer-day cost is used as the primary break-even reference because ATB works each customer through multiple call attempts on average.

all rights reserved

prometeia

Internal And Trusted Partner Use Only Document owned by Prometeia

Action Uplift methodology

Uplift is treated as a scenario-based assumption to determine when an action becomes economically justified

Break-even ERA = Cost / Uplift

Expected Benefit = ERA x Uplift should be higher than Cost -> ERA x Uplift > Cost -> Break Even ERA = Cost / Uplift

Action uplift represents the assumed relative reduction in the customer-level expected roll amount resulting from the collection action. It is used as a scenario-based assumption to estimate the break-even expected roll amount for each channel; it does not represent an observed or guaranteed impact. Action uplift should be validated and refined through observed treatment outcomes.

Action channel	Cost basis	Unit cost	Conservative uplift / break-even	Base uplift / break-even	Upside uplift / break-even
Reminder SMS	Per SMS	**0.031 AZN**	0.1% / **31 AZN**	0.3% / **10 AZN**	0.5% / **6 AZN**
Robo call / IVR	Per attempt	**0.20 AZN**	0.5% / **40 AZN**	1.0% / **20 AZN**	2.5% / **8 AZN**
Agent Call	Per customer-day	**1.75 AZN**	2.5% / **70 AZN**	5.0% / **35 AZN**	7.5% / **23 AZN**

The action matrix is calibrated by comparing the estimated cost of each treatment channel with its expected uplift on the customer-level expected roll amount. Lower-cost channels such as SMS and robo call can be economically justified at lower risk levels, while manual calls should be reserved for customers with higher expected roll amount.

all rights reserved

Expected Roll Amount (ERA) Distribution

The full 1–30 DPD population is ranked on a common economic risk metric, enabling cross-segment treatment decisions

Expected Roll Amount = Roll Probability × Total Exposure

Different DPD bands use different probability assignment methods, but all customers are converted into the same monetary risk metric.

Probability sources

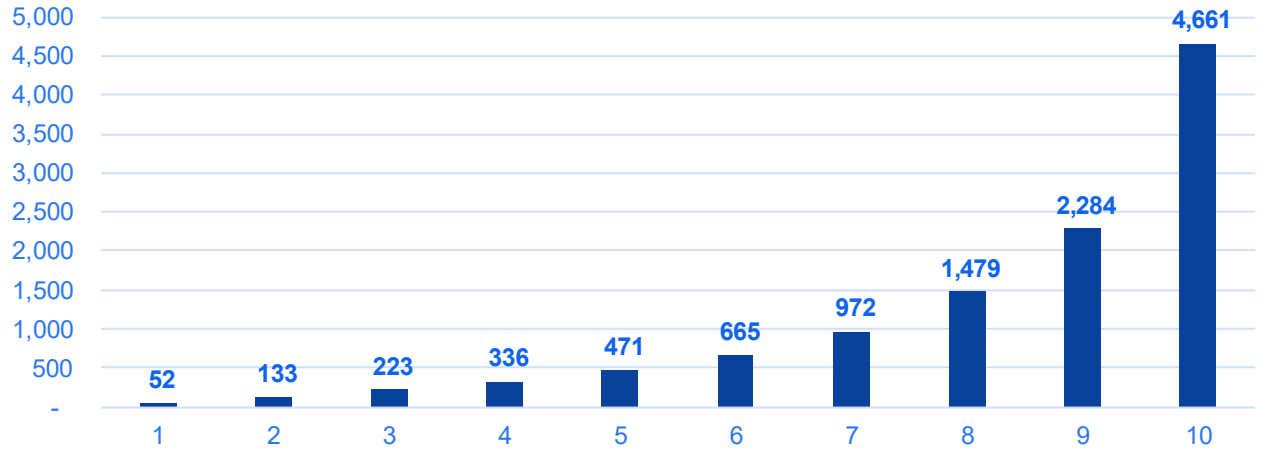
- 1–3 DPD: simple rule-based probability using max DPD within the last 3 months
- 4–10 DPD: model-based roll probability
- 11–30 DPD: DPD-bucket roll rate

Capacity Logic

- Team capacity: **6 agents × 80 successful contacts = 480 successful contacts/day**
- Senior agents: focus on **Bucket 10 and complex high-risk cases**
- Regular agents: focus on **Buckets 9 and 8**
- Bucket 7 is called only if capacity remains
- Buckets 1–6 are managed mainly through digital reminders

Expected Roll Amount Percentile Distribution

(daily average 200 customer per bucket)



Starting from the lowest risk class, the call action always economically feasible

Risk / Action Classification Matrix

Risk Segment	Buckets	Main Treatment
Low	1–4	SMS or email
Medium	5–6	SMS + robo call + email
Medium-High	7	Robo call + selective agent call
High	8	Agent call
Very High	9	Agent call priority
Critical	10	Senior agent priority

prometeia

all rights reserved

Internal And Trusted Partner Use Only Document owned by Prometeia

Delinquency Distribution by ERA Bucket

The below analysis ensures that the high-risk buckets (8, 9, 10) not only consist of clients with high DPD thanks to granular approach adopted based on decision tree model

	DPD_BUCKET					
ERA BUCKET	**DPD 1 - 3**	**DPD 4 - 10**	**DPD 11-16**	**DPD 17 - 23**	**DPD 24 - 30**	**TOTAL**
1	77.9%	16.6%	3.3%	1.5%	0.7%	100.0%
2	69.4%	22.2%	4.5%	2.6%	1.3%	100.0%
3	64.4%	24.9%	5.5%	3.6%	1.6%	100.0%
4	62.2%	26.0%	6.3%	3.6%	1.9%	100.0%
5	53.8%	29.6%	9.0%	5.2%	2.4%	100.0%
6	36.1%	37.3%	12.8%	9.4%	4.5%	100.0%
7	20.1%	40.2%	19.5%	12.4%	7.8%	100.0%
8	11.4%	36.1%	24.5%	16.4%	11.6%	100.0%
9	8.7%	31.1%	24.3%	21.6%	14.3%	100.0%
10	6.9%	24.1%	14.8%	25.6%	28.6%	100.0%

all rights reserved

prometeia

Internal And Trusted Partner Use Only Document owned by Prometeia

Risk Based Action Matrix

Risk Segment	Bucket	Primary Action	Supporting Action	Agent Assignment
Low	1–4	SMS + email	No	No agent call
Medium	5–6	Robo call	SMS or email	No agent call
Medium-High	7	Robo call + SMS	Agent call if capacity remains	Regular agent if capacity remains
High	8	Agent call	SMS before call + email after call	Regular agent
Very High	9	Agent call	SMS before call + email after call	Regular agent priority
Critical	10	Senior agent call	SMS before call + email after call	Senior agent priority

all rights reserved

prometeia

Internal And Trusted Partner Use Only Document owned by Prometeia

If the contact is successful;

Successful contact triggers a minimum 5-day agent call cooldown period, while low-intensity digital reminders may continue based on the agreed payment date or delinquency status.

Situation	Action
Successful contact established	No agent call for 5 days
Promise to pay obtained	Payment date is tracked
Promise-to-pay date is approaching	SMS / email reminder may be sent
No payment within 5 days	Customer is re-included in the call list based on risk segment

all rights reserved

prometeia

Internal And Trusted Partner Use Only Document owned by Prometeia

Unsuccessful call retry mechanism

Risk Segment	After First Unsuccessful Call Attempt
Bucket 10	Retry at a different time on the same day + retry again the next day
Bucket 9	Retry at a different time on the same day + retry again the next day
Bucket 8	Retry after 1–2 days
Bucket 7	Retry after 2–3 days or subject to capacity availability
Buckets 1–6	No agent call

all rights reserved

prometeia

Internal And Trusted Partner Use Only Document owned by Prometeia

Current operational limitations & recommendations

Current operational limitations

- No IVR capability and no email collection action in place
- Dialer usage is limited to queue display and anonymous outbound calls
- Customer reachability is tracked manually in Excel
- No standardized collection performance reporting pack

Bank actions required

1. Robo call (IVR)
Use SMS + structured agent calling as Phase 1. Enable robo call if the Bank can operationally support it.

2. Upgrade dialer usage
Feed risk-ranked daily call lists into the dialer, Replace Excel-only tracking with a structured outcome table covering RPC, no answer, busy, unreachable, wrong number, payment promise, and already paid..

3. Build monitoring pack
Create daily and weekly reporting for attempt rate, RPC, successful contact, cure, roll-forward, payment per contact, and agent productivity.

4. Implement contact rules
Successful contact triggers minimum 5-day suppression for agent calls. Unsuccessful calls do not trigger suppression; retry depends on risk and failure reason.

prometeia

all rights reserved

Internal And Trusted Partner Use Only Document owned by Prometeia

AGENDA

SME Credit Risk Scorecard

all rights reserved

prometeia

Internal And Restricted Use Only Document © Prometeia

SME Credit Risk Scorecard

To assess the effectiveness of the current Small (Kicik) and Medium (İri) voting algorithm and identify potential enhancement opportunities, the existing scorecards were backtested against historical default behaviour and bureau rating information.

Target & Sample Construction

- Customers having AKB's Worst Rating either at the credit approval date or at the end of the performance period, or having 30+ DPD delinquency or entering NPL status during the performance period, were classified as Bad observations. All remaining observations were classified as Good.
- The development sample was unified such that each customer was represented by a single observation per month, with multiple applications within the same month consolidated into one record.
- The final validation sample consists of 221 observations within the scope of the Small Voting Algorithm and 34 observations within the scope of the Medium Voting Algorithm, with observed Bad Rates of 24% and 47%, respectively.
- GINI analysis was performed at overall score, category and question levels to assess scorecard effectiveness and identify enhancement opportunities.

Form Name	Count	Bad Count	Bad %
Kicik (Small)	221	54	24%
İri (Medium)	34	16	47%

Current Weights of the Voting Algorithms

Category	Question	Weight	Max Category Score
1.Credit History Analysis	1.2. Obtaining Credit Bureau Reports for Related Parties	35%	20%
	1.3. Unjustified Recent Credit Applications in Credit Bureau Records	5%	
	1.4. Unjustified 0–30 Day Delinquencies on Banking Obligations	10%	
	1.5. Unjustified 30+ Day Delinquencies on Banking Obligations	20%	
	1.6. Repayment of Loans Through Monthly Installments	15%	
	1.7. Increase in debt burden	15%	
2.Business Analysis	2.1. Business Relationship to the Applicant and Other Considerations	33%	20%
	2.2. Business structure and management expertise	33%	
	2.3. Documentation and Reporting	33%	
3.Financial Analysis	3.1. Balance Sheet	20%	35%
	3.2. Income Statement	20%	
	3.3. Cash Flows	20%	
	3.4. Financial Statement Comparison	20%	
	3.5. Financial Ratios	20%	
4.Loan Purpose Analysis	4.1. Availability of Supporting Documents for Loan Purpose	25%	15%
	4.2. Loan Purpose Efficiency	50%	
	4.3. Availability of monitoring capabilities for the intended loan purpose	25%	
5.Colleteral Analysis	5.1. Ownership of Real Estate Collateral by Business Shareholders or Close Family Members.	50%	10%
	5.2. Risk Level of Real Estate Collateral	20%	
	5.3. Suitability of Guarantor(s)	30%	

prometeia

Internal And Trusted Partner Use Only Document owned by Prometeia

SME Credit Risk Scorecard – Small Scorecard

1 Assessment of current scorecard performance

- Current Small (Kicik) scorecard has an overall **GINI of 0.05,** indicating very limited discriminatory power

.

- Category and question level analyses were performed to identify the key performance drivers.

2 Category & Question Level Validation

3 Identification of predictive components

- **The first section, Credit History Analysis**, demonstrated the strongest discriminatory power among all categories. All questions were found to be economically significant and retained for further development, except the question assessing recent unjustified credit applications, which did not demonstrate sufficient predictive power.
- **Section 2, Business Analysis,** exhibited a lack of discriminatory power overall. Only the second question which is assessing Business Structure and Management Expertise demonstrated sufficient discriminatory ability and economic significance. Therefore, this was the only question retained from this section for further development.
- **Financial Analysis** did not demonstrate the expected risk relationship and was characterized by weak monotonicity and highly concentrated distributions across categories.
- **Loan Purpose Analysis and Collateral Analysis** primarily reflected transaction-specific characteristics rather than obligor creditworthiness and did not demonstrate meaningful predictive power.

4 Review of score allocation and question weights optimization

•The original section-based weighting structure was replaced with a revised question-level weighting approach. The final weighting configuration was selected based on the overall performance and interpretability of the scorecard.

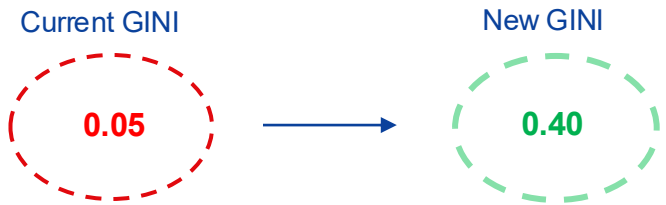
• The optimized scorecard achieved a **GINI of 0.40**, representing a significant improvement over the current scorecard structure.

all rights reserved

Categories	GINI	*Economic Significance
SECTION_1_CREDIT_HISTORY_ANALYSIS	0.31	TRUE
SECTION_2_BUSINESS_ANALYSIS	-0.02	FALSE
SECTION_3_FINANCIAL_ANALYSIS	-0.06	FALSE
SECTION_4_LOAN_PURPOSE_ANALYSIS	-0.20	FALSE
SECTION_5_COLLATERAL_ANALYSIS	-0.12	FALSE

**Economic Significance indicates whether the relationship between the category score and default risk is directionally consistent with expectations, i.e., higher scores are associated with lower default rates.*

Final Form Questions	New Weight	GINI	IV
Q1.2 RELATED_PARTY_CREDIT_BUREAU_CHEK	**5%**	0.03	0.06
Q1.4 UNJUSTIFIED_0_30_DPD	**25%**	0.2	0.16
Q1.5 UNJUSTIFIED_30PLUS_DPD	**20%**	0.19	0.58
Q1.6 MONTHLY_INSTALLMENT_REPAYMENT	**10%**	0.13	0.09
Q1.7 INCREASE_IN_DEBT_BURDEN	**20%**	0.18	0.15
Q2.2 MANAGEMENT_AND_GOVERNANCE	**20%**	0.19	0.29



prometeia

Internal And Trusted Partner Use Only Document owned by Prometeia

SME Credit Risk Scorecard – Small Scorecard

- The fine-tuned scorecard consists of six questions that demonstrated predictive power and economic significance during the redevelopment process. Each question is assigned a weight reflecting its relative contribution to risk differentiation.
- For each question, responses are mapped to a predefined score representing the associated risk level. The final score is then calculated as the weighted sum of the individual question scores, resulting in a single overall risk score ranging from 0 to 100.
- Higher scores indicate a more favorable risk profile, while lower scores indicate higher credit risk.
- Rating scale and cut off analysis will be provided later.

New Weights

No.	Question	Answer	Answer Score	Weight
1	Obtaining Credit Bureau Reports for Related Parties	Yes	1	5%
		No	0	
2	Unjustified 0–30 Day Delinquencies on Banking Obligations	No	1	25%
		Yes	0	
3	Unjustified 30+ Day Delinquencies on Banking Obligations	No	1	20%
		Yes	0	
4	Repayment of Loans Through Monthly Installments	Yes	1	10%
		No	0	
5	Increase in debt burden	No	1	20%
		Yes	0	
6	Business structure and management expertise	Low Risk	0	20%
		High Risk	1	

Final Score = Σ (Question Score × Weight)

Score Distribution

Score Band	Obs	Bad	Bad Rate	Obs_%	Bad_%
0-20	4	4	100.0%	1.8%	7.4%
21-40	16	13	81.3%	7.2%	24.1%
41-60	56	13	23.2%	25.3%	24.1%
61-80	108	19	17.6%	48.9%	35.2%
81-100	37	5	13.5%	16.7%	9.3%

all rights reserved

prometeia

Internal And Trusted Partner Use Only Document owned by Prometeia

SME Credit Risk Scorecard – Medium Scorecard

1 Assessment of current scorecard performance

- Current İri scorecard has an overall **GINI of 0.01,** indicating lack of discriminatory power

.

- Category and question level analyses were performed to identify the key performance drivers.

2 Category & Question Level Validation

3 Identification of predictive components

- **The Credit History Analysis and Business Analysis** sections demonstrated predictive power and were retained as aggregated section-level components in the final scorecard.
- **For the Financial Analysis section**, the strongest-performing questions were selected individually and incorporated into the final score calculation. The final weighting structure was then assigned across the retained sections and financial assessment questions.
- **Loan Purpose Analysis and Collateral Analysis** primarily reflected transaction-specific characteristics rather than the underlying creditworthiness of the borrower. Both sections demonstrated weak and inconsistent discriminatory power and were therefore excluded from the revised scorecard framework.

4 Review of score allocation and question weights optimization

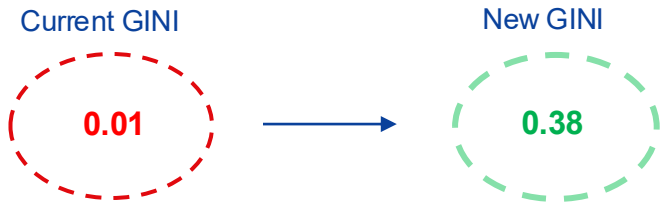
• The original weighting structure was revised and alternative weighting configurations were assessed. The final weighting configuration was selected based on the overall performance and interpretability of the scorecard.

• The optimized scorecard achieved a **GINI of 0.38** representing a significant improvement over the current scorecard structure.

Categories	GINI	*Economic Significance
SECTION_1_CREDIT_HISTORY_ANALYSIS	0.13	TRUE
SECTION_2_BUSINESS_ANALYSIS	0.38	TRUE
Q_3_1_BALANCE_SHEET	0.28	TRUE
Q_3_2_INCOME_STATEMENT	0.22	TRUE
Q_3_4_FINANCIAL_STATEMENT_COMPARISON	0.13	TRUE
Q_3_5_FINANCIAL_RATIOS	0.11	TRUE
SECTION_4_LOAN_PURPOSE_ANALYSIS	-0.03	FALSE
SECTION_5_COLLATERAL_ANALYSIS	-0.50	FALSE

**Economic Significance indicates whether the relationship between the category score and default risk is directionally consistent with expectations, i.e., higher scores are associated with lower default rates.*

Final Form Questions	New Weight	GINI
SECTION_1_CREDIT_HISTORY_ANALYSIS	30%	-0.1215
SECTION_2_BUSINESS_ANALYSIS	30%	-0.3785
Q_3_1_BALANCE_SHEET	15%	-0.2813
Q_3_2_INCOME_STATEMENT	15%	-0.2292
Q_3_4_FINANCIAL_STATEMENT_COMPARISON	5%	-0.1319
Q_3_5_FINANCIAL_RATIOS	5%	-0.1146



prometeia

all rights reserved

Internal And Trusted Partner Use Only Document owned by Prometeia

SME Credit Risk Scorecard – Medium Scorecard

- The final scorecard consists of six components, including two section-level assessments and four financial assessment questions. Each component is assigned a weight reflecting its relative importance in the overall score calculation. Component scores are first converted to a standardized scale and then multiplied by their respective weights. The final score is obtained by summing the weighted component scores, resulting in a value between 0 and 100.

Score Distribution

Score Band	Obs	Bad	Obs_%	Bad Rate	Bad_%
0-20	0	0	0.00%	0.00%	0.00%
21-40	1	1	2.94%	100.00%	6.25%
41-60	14	8	41.18%	57.14%	50.00%
61-80	19	7	55.88%	36.84%	43.75%
81-100	0	0	0.00%	0.00%	0.00%

New Weights of the Voting Algorithms

No	Question	Answer	Score	Weight
1	SECTION_1_CREDIT_HISTORY_ANALYSIS	Section Score (from 0 to 20)	Section Score / 20	30%
2	SECTION_2_BUSINESS_ANALYSIS	Section Score (from 0 to 20)	Section Score / 20	30%
3	BALANCE_SHEET	Low Risk	1	15%
		Low-Medium Risk	0.75	
		Moderate Risk	0.5	
		Medium-High Risk	0.25	
		High Risk	0	
4	INCOME_STATEMENT	Low Risk	1	15%
		Low-Medium Risk	0.75	
		Moderate Risk	0.5	
		Medium-High Risk	0.25	
		High Risk	0	
5	FINANCIAL_STATEMENT_COMPARISON	Low Risk	1	5%
		Low-Medium Risk	0.75	
		Moderate Risk	0.5	
		Medium-High Risk	0.25	
		High Risk	0	
6	FINANCIAL_RATIOS	Low Risk	1	5%
		Low-Medium Risk	0.75	
		Moderate Risk	0.5	
		Medium-High Risk	0.25	
		High Risk	0	

all rights reserved

prometeia

Internal And Trusted Partner Use Only Document owned by Prometeia

Next Steps – SME Scorecard

1. Fine tunings finalization; develop alternative quanitative financial scorecard based on the financial ratios instead of qualitative assessment of the financials.
2. Rating Scale construction
3. Knock out and override rules
4. Cut off analysis – auto reject criteria

all rights reserved

prometeia

Internal And Trusted Partner Use Only Document owned by Prometeia